Real Numbers & The Pythagorean Theorem: Radicals & Square Roots

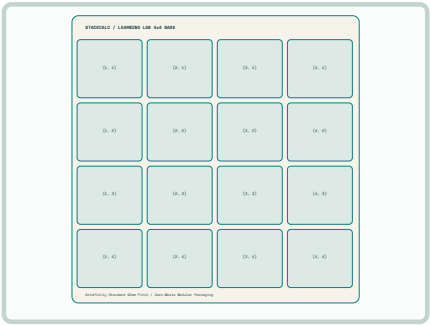
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.8.G.B.7

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Gridfinity Baseplate

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Form right triangle with legs 3 and 4 along grid studs; hypotenuse span is 5 units.

Sketch Model on Paper:
Draw squares on each side: 3^2 (9) + 4^2 (16) = 5^2 (25).

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

1.	6	X: 6	Leg a = 6 into X
2.	x^2	X: 36	Square a: 6^2 = 36
3.	8	Y: 36 | X: 8	Leg b = 8 into X
4.	x^2	Y: 36 | X: 64	Square b: 8^2 = 64

EXPECTED LCD X: c = 10

1 **Find hypotenuse c for legs a = 5, b = 12.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Find hypotenuse c for legs a = 9, b = 12.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Find leg a if hypotenuse c = 25 and leg b = 24: $\sqrt{(25^2 - 24^2)}$.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Approximate $\sqrt{50}$ to two decimal places.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 7

Real Numbers & The Pythagorean Theorem: Radicals & Square Roots • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Gridfinity 4x4 Coordinate Baseplate & Triangle

CAD Blueprint: gridfinity_baseplate

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/scad/lib/gridfinity.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Charcoal 4x4 Baseplate & Teal Cubes (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Find hypotenuse c for legs a = 5, b = 12.** Standard: CCSS.MATH.8.G.B.7	c = 13	[5] [x²] [1][2] [x²] [+] [√x]
2	**Find hypotenuse c for legs a = 9, b = 12.** Standard: CCSS.MATH.8.G.B.7	c = 15	[9] [x²] [1][2] [x²] [+] [√x]
3	**Find leg a if hypotenuse c = 25 and leg b =** Standard: CCSS.MATH.8.G.B.7	a = 7	[2][5] [x²] [2][4] [x²] [-] [√x]
4	**Approximate √50 to two decimal places.** Standard: CCSS.MATH.8.G.B.7	7.07	[5][0] [√x]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for forgetting to take the square root at the end of $a^2 + b^2$.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.